HW3_MasonHolcombe

Mason Holcombe

Chapter 4

3)

a) Covariance = 2 and does not 0, not independent

b) Covariance = 0, so X2 and X3 are independent

c) Cov(x1,x3) = 0, cov(x2, x3) = 0, so (X1,X2) and X2 are independent

d) 1/2 * (cov(x1,x3) and cov(x2, x3)). Both of these are 0, so 1/2 * 0 = 0. D is independent

e) cov(x2,x2) = 5, cov(-5/2x1, x2) = -5/2 * cov(x1,x2) = -5/2 * -2 = 5, and cov(x2, x3) = 0. So the final covariance is 5 + 5 + 0 = 10 and this does not equal 0. So x2 and (e) are not independent.

4)

a)

```
mu = c(2,-3,1)
sigma = matrix(c(1,1,1,
                 1,3,2,
                 1,2,2), byrow=T, ncol=3)
a = c(3, -2, 1)
```

```
mean = t(a) %*% mu
var = t(a) %*% sigma %*% a
c(mean, var)
```

```
[1] 13  9
```

3x1 - 2x2 + x2 ~ N(13, 9)

19)

a) ~ χ^2_6 chi2 with 6 df

b)

- p-variate normal with mu as mean and 1/20 * sigma as covariance matrix
- p-variate normal with 0 as mean and sigma as covariance matrix

c) wishart distribution with 19 (20-1) df and sigma as the scale matrix

29)

```
air = read.table("../johnson_wichern_data/T1-5.DAT")
x5 = air["V5"]
x6 = air["V6"]

X = cbind(x5, x6)
n = nrow(X)
```

a)

```
xbar = colMeans(X)
S = cov(X)
Sinv = solve(S)

result = numeric(n)
for (i in 1:n){
  xi = matrix(as.numeric(X[i, ]),
              ncol=1)
  result[i] = t(xi - xbar) %*% Sinv %*% (xi - xbar)
}
(result)
```

```
 [1]  0.4606524  0.6592206  2.3770610  1.6282902  0.4135364  0.4760726
 [7]  1.1848895 10.6391792  0.1388339  0.8162468  1.3566301  0.6228096
[13]  5.6494392  0.3159498  0.4135364  0.1224973  0.8987982  4.7646873
[19]  3.0089122  0.6592206  2.7741416  1.0360061  0.7874152  3.4437748
[25]  6.1488606  1.0360061  0.1388339  0.8856041  0.1379719  2.2488867
[31]  0.1901188  0.4606524  1.1471939  7.0857237  1.4584229  0.1224973
[37]  1.8984708  2.7782596  8.4730649  0.6370218  0.7032485  1.8013611
```

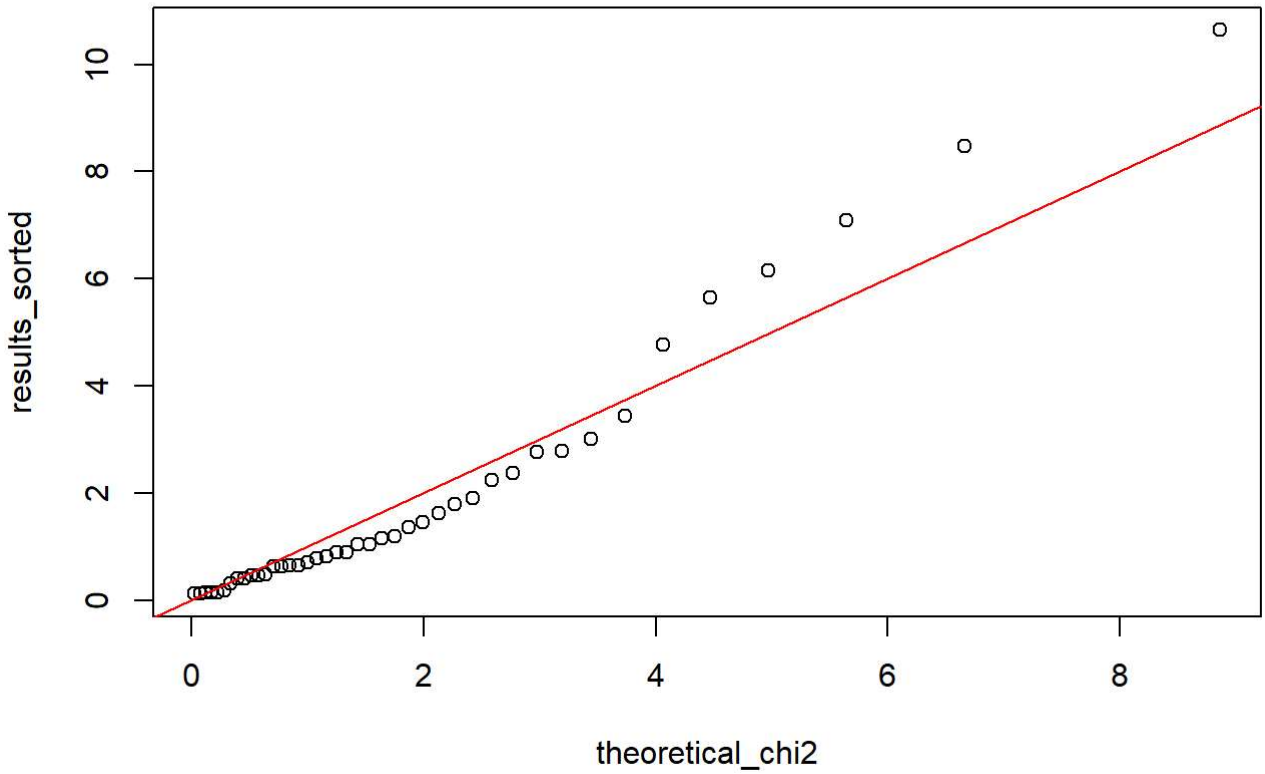
b)

```
cutoff = qchisq(0.5, df=2)
mean(result <= cutoff)
```

```
[1] 0.6190476
```

c)

```
results_sorted = sort(result)
theoretical_chi2 = qchisq((1:n - (1/2)) / n, df=2)

plot(theoretical_chi2, results_sorted)
abline(0,1,col='red')
```



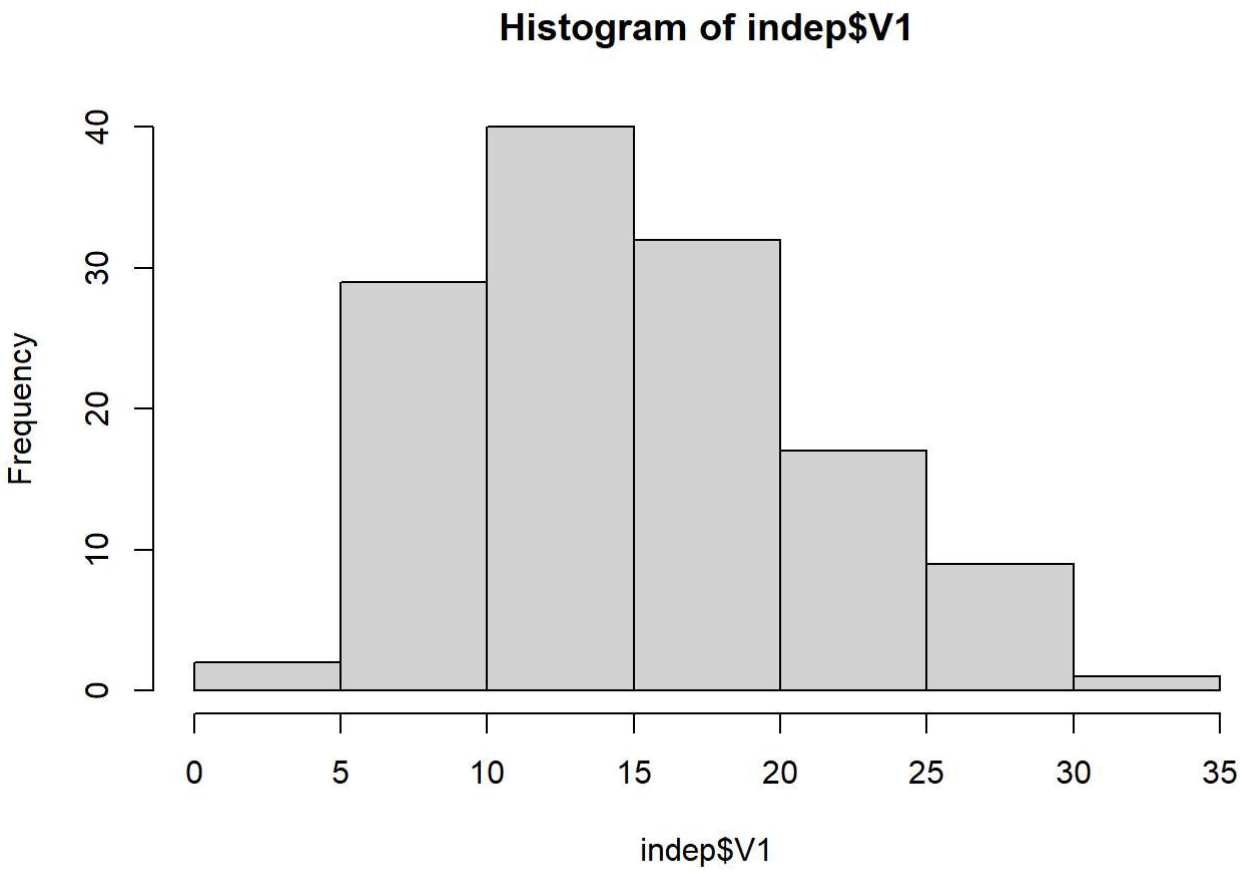
39)

```
data39 = read.table("../johnson_wichern_data/T4-6.DAT")
indep = data39["V1"]
supp = data39["V2"]
benev = data39["V3"]
conform = data39["V4"]
leader = data39["V5"]

gender = data39["V6"]
socio = data39["V7"]
```

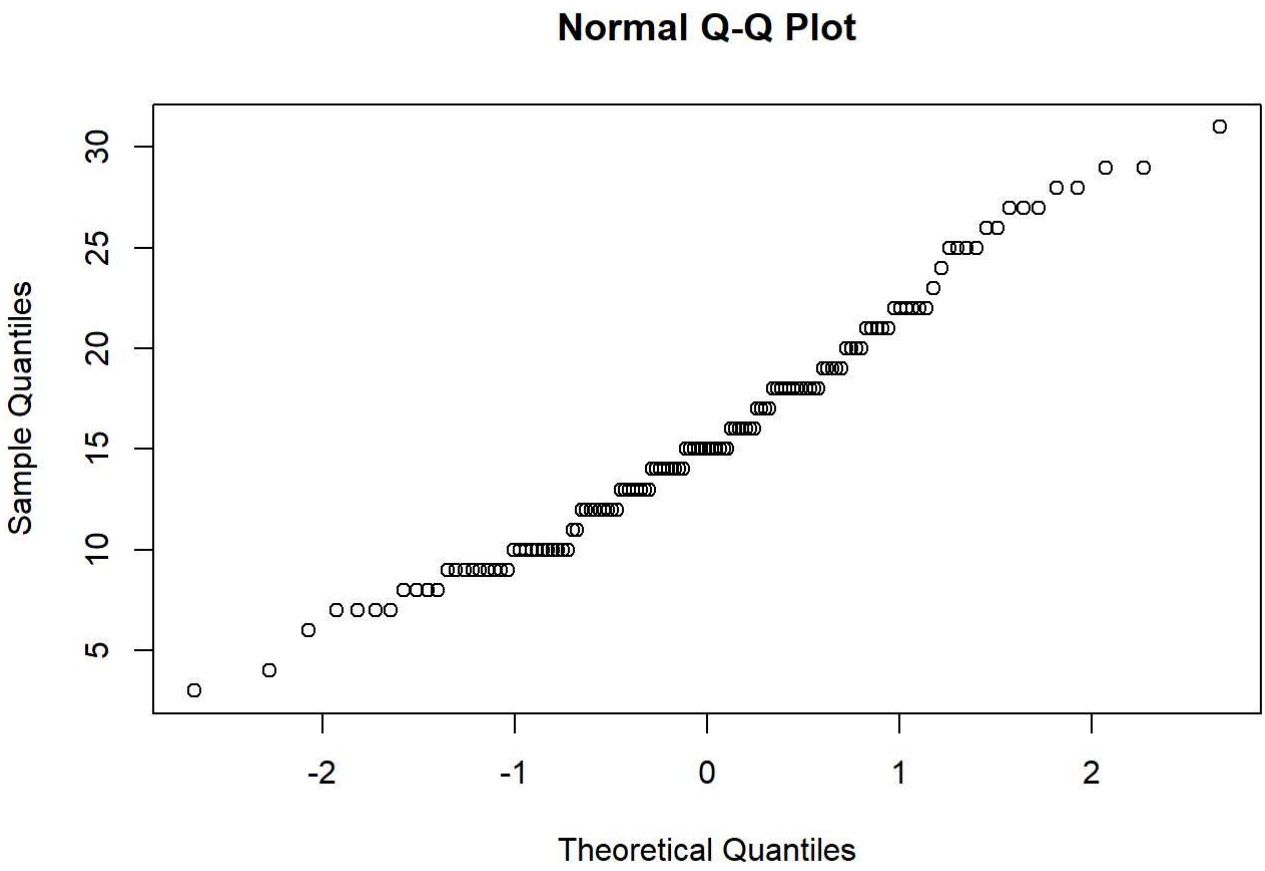
a)

Independence - NORMAL

```
hist(indep$V1)
```

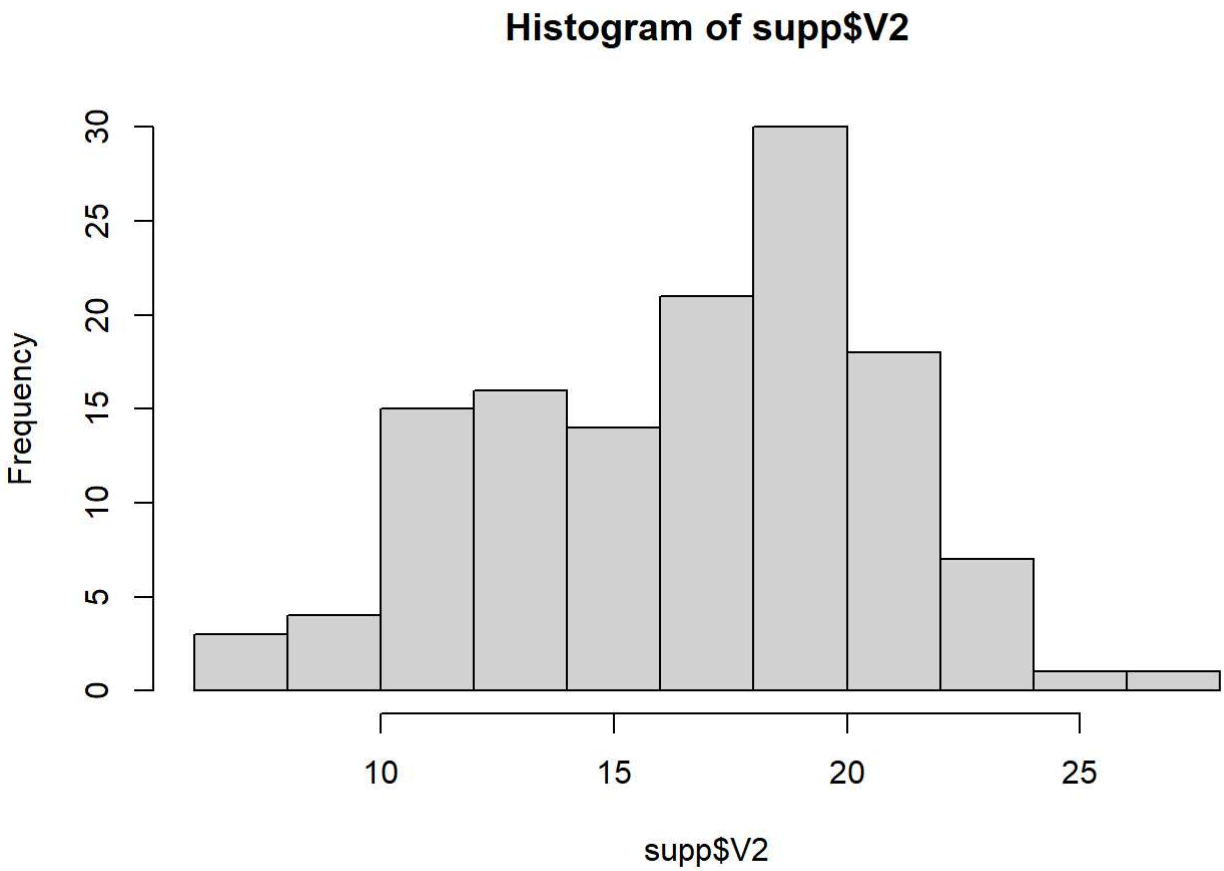



```
qqnorm(indep$V1)
```

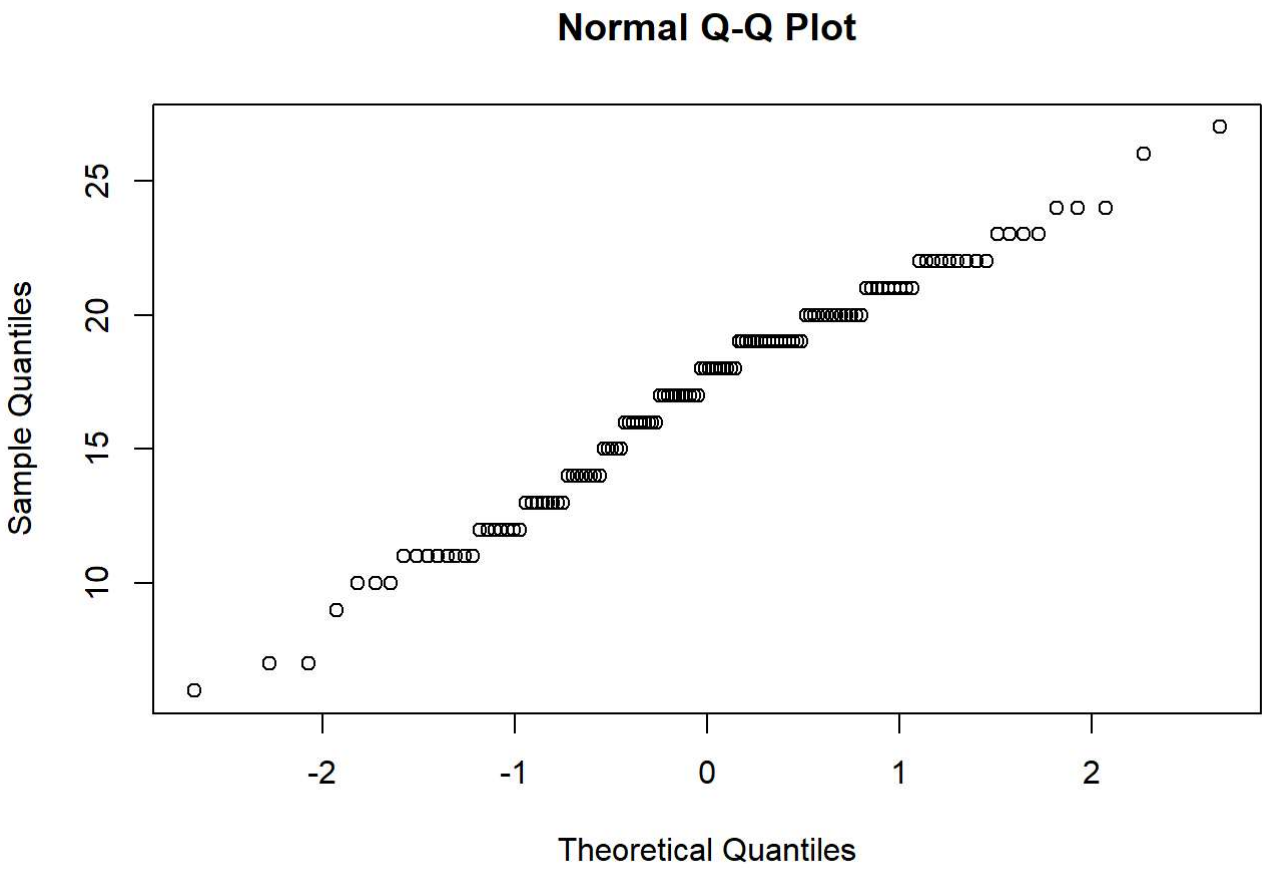


Support – NOT NORMAL

```
hist(supp$V2)
```

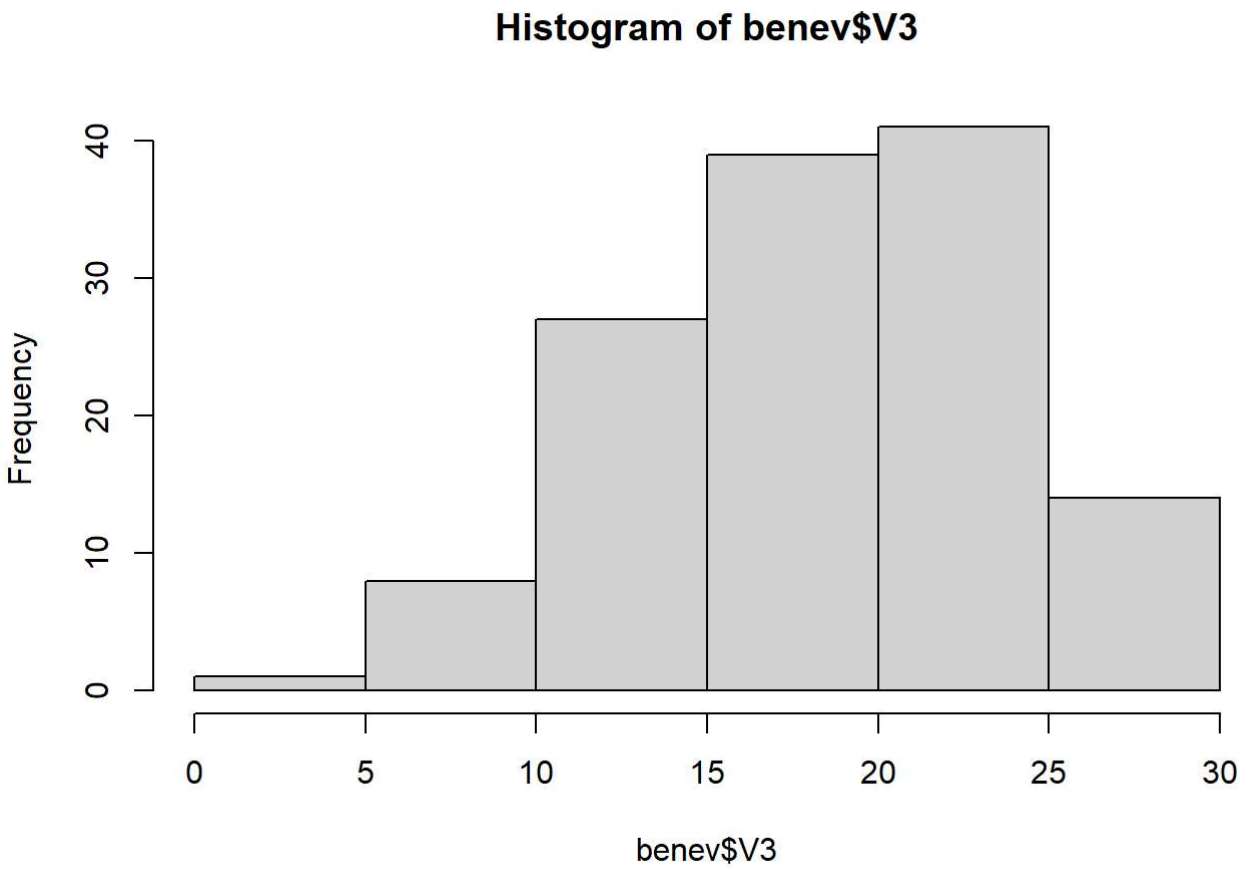



```
qqnorm(supp$V2)
```

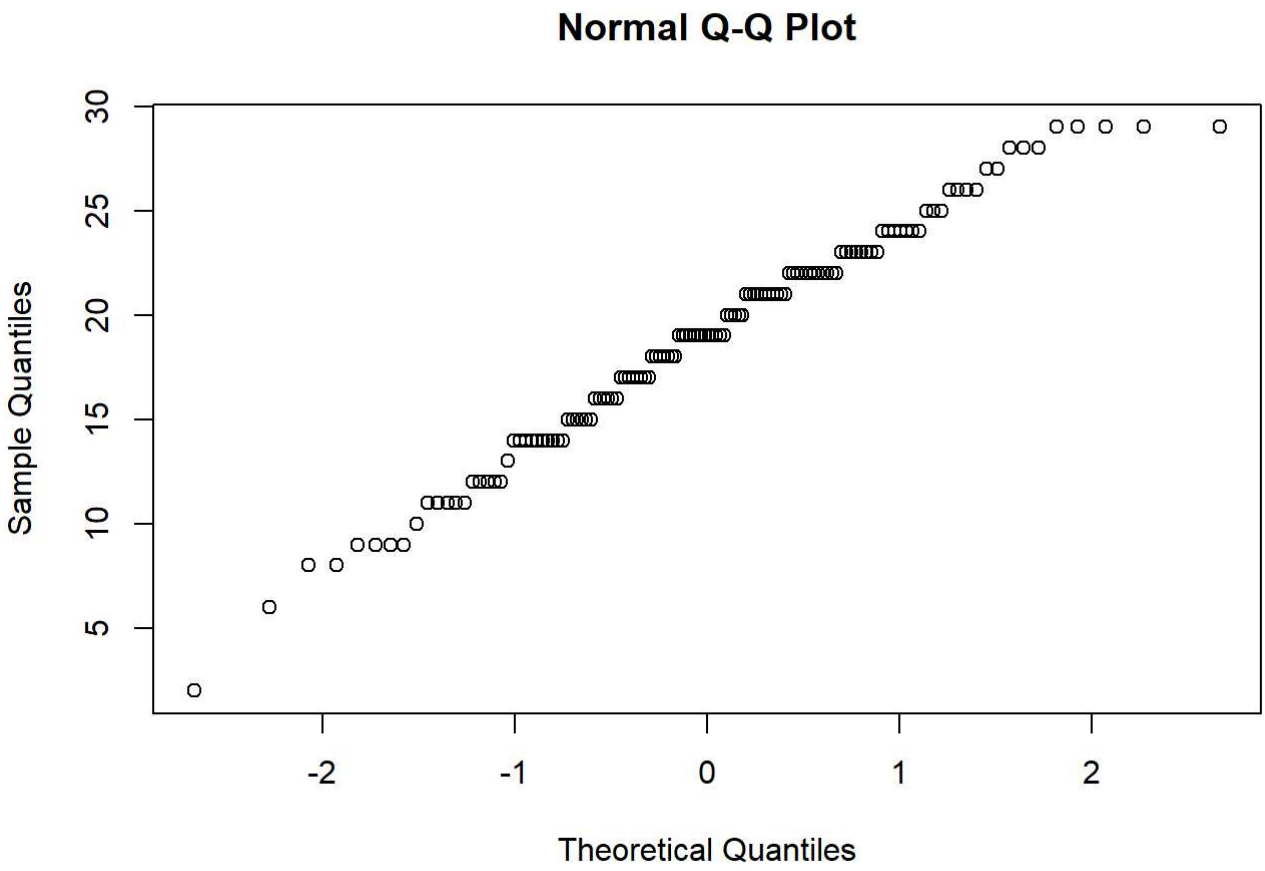


Benevolence – NORMAL

```
hist(benev$V3)
```

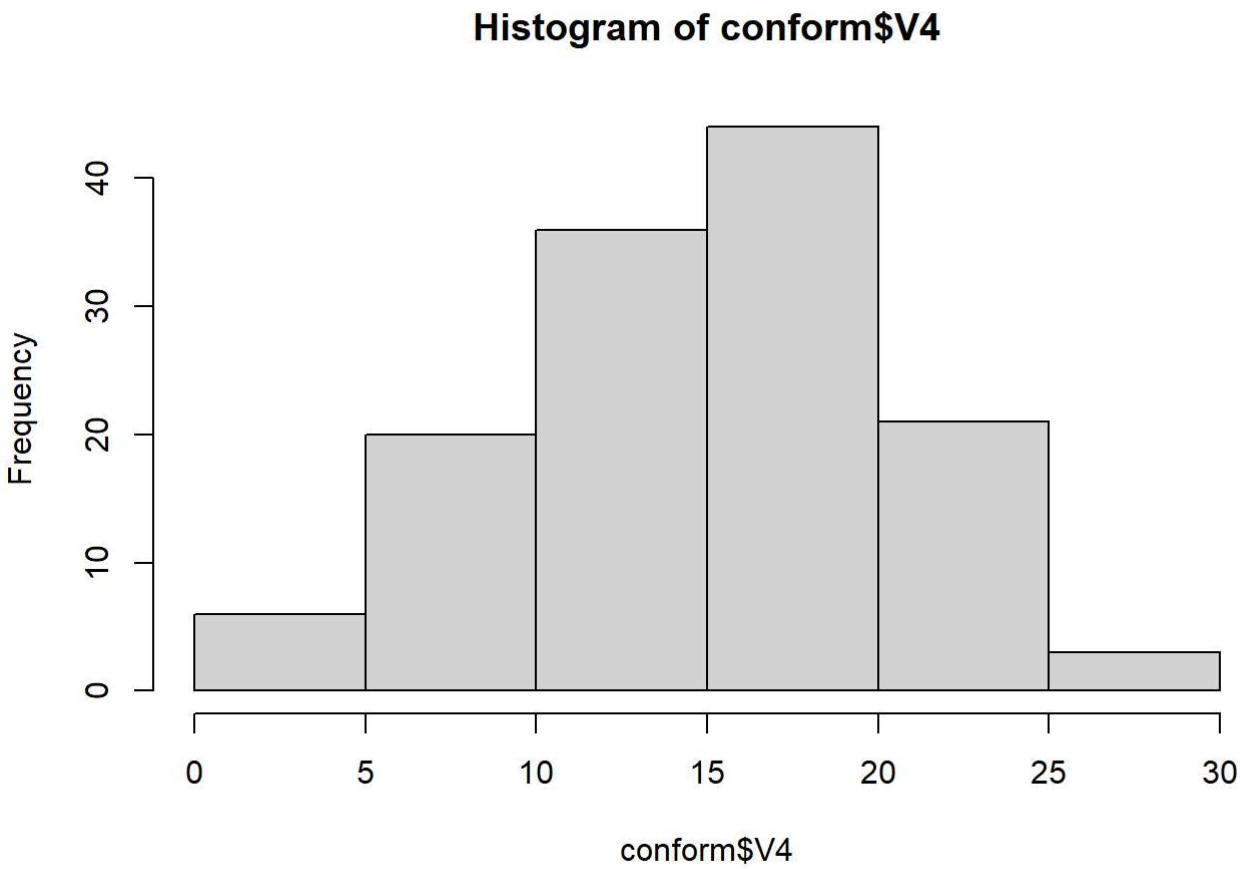



```
qqnorm(benev$V3)
```

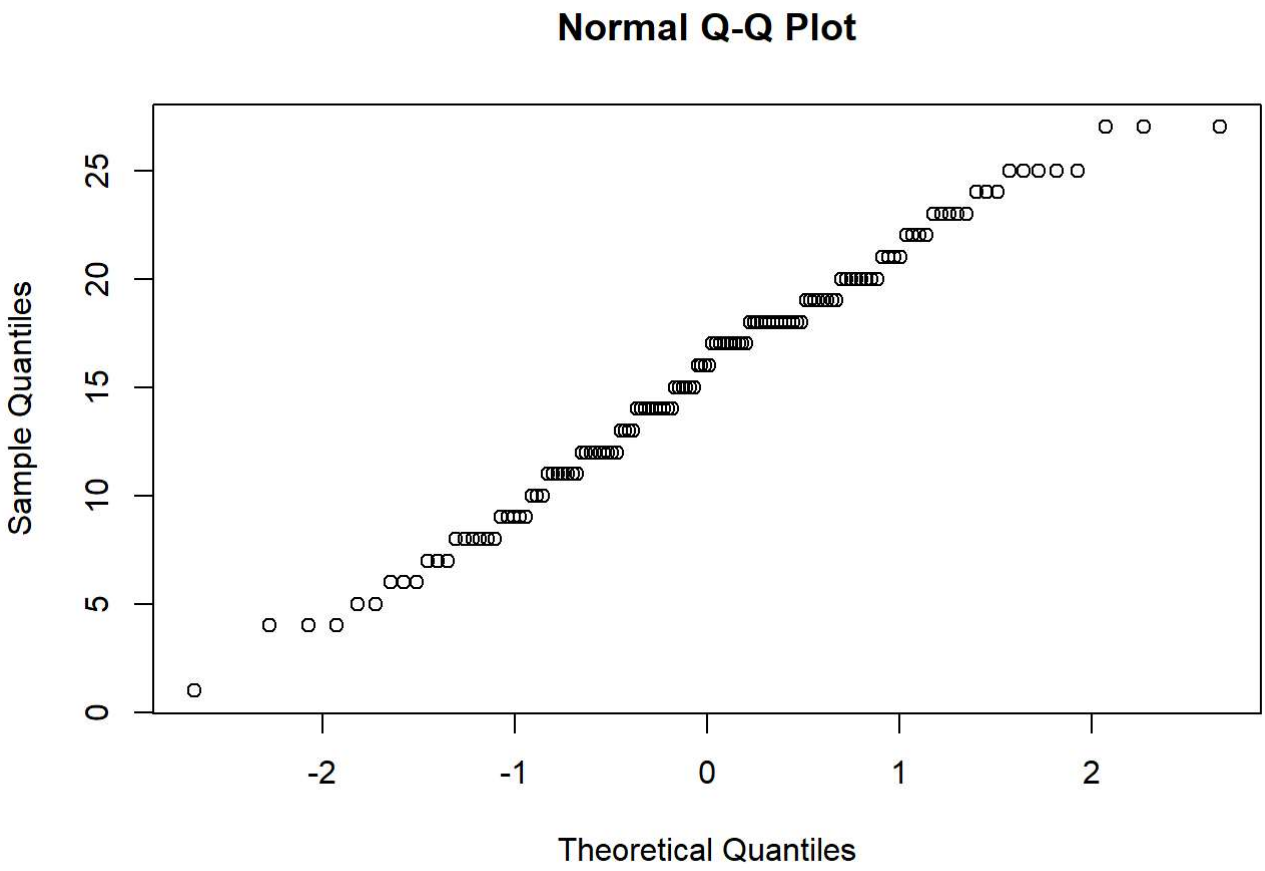


Conform - NORMAL

```
hist(conform$V4)
```

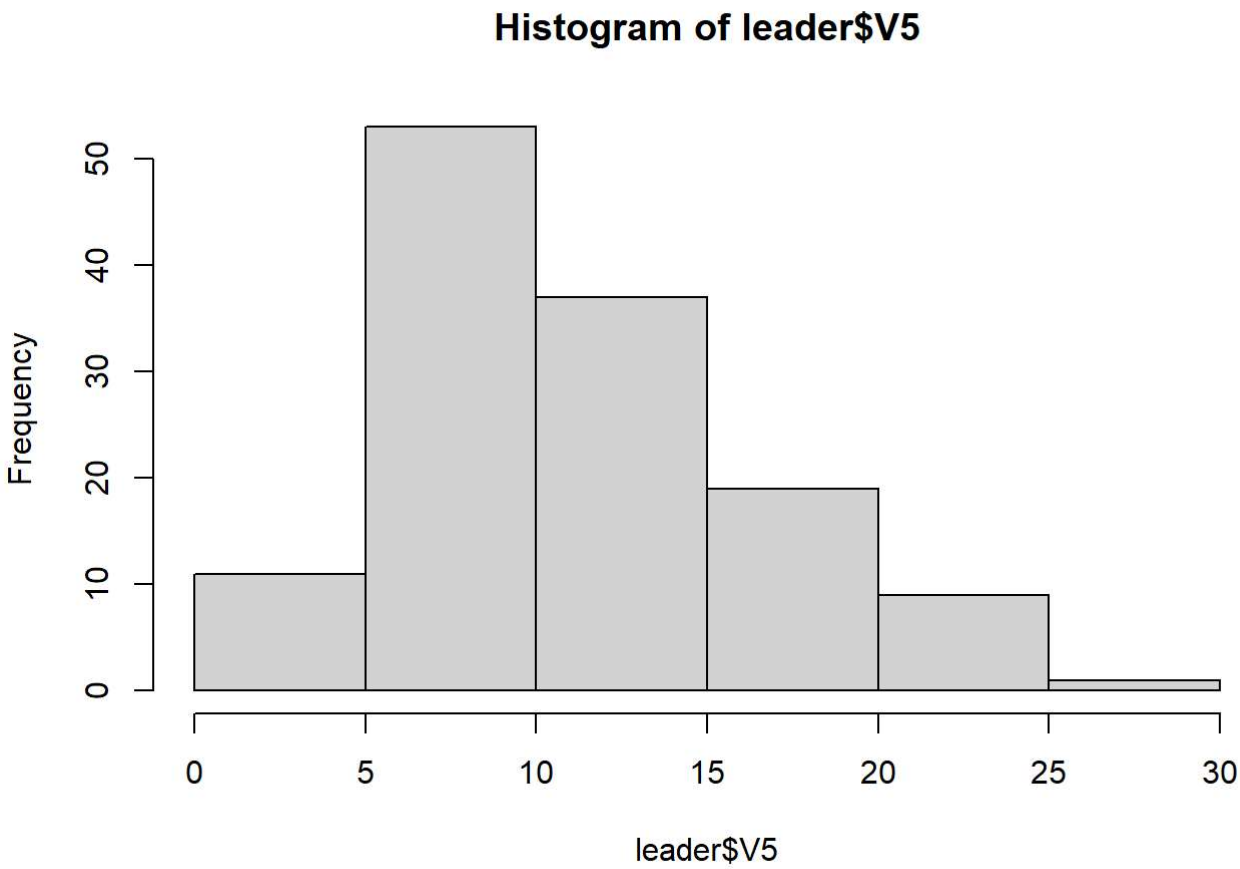



```
qqnorm(conform$V4)
```

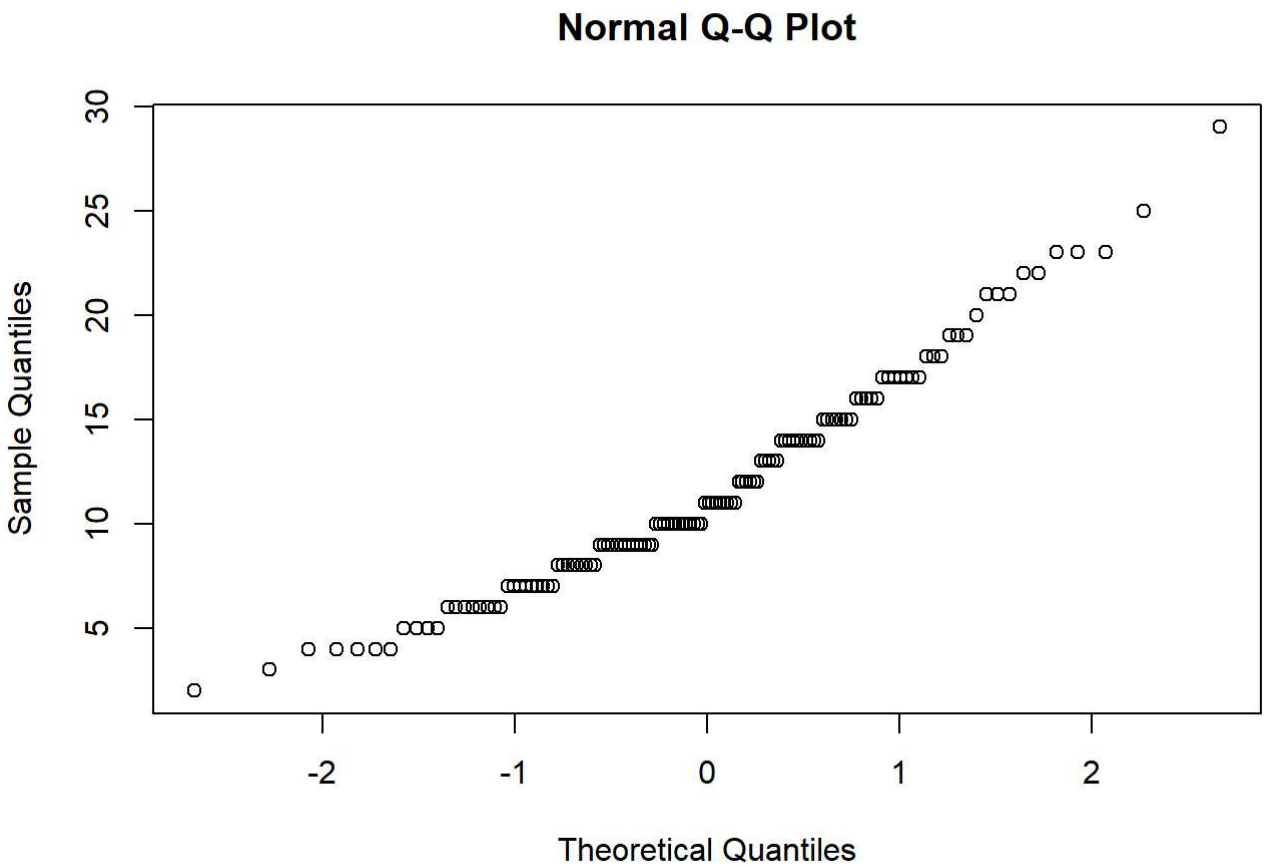


Leader – NOT NORMAL (SKEWED RIGHT)

```
hist(leader$V5)
```




```
qqnorm(leader$V5)
```



b)

```
varnames = c("V1", "V2", "V3", "V4", "V5")
features = data39[ ,varnames]
```

```
library("MVN")
```

```
Warning: package 'MVN' was built under R version 4.5.2
```

```
mvn_result = mvn(data = features, mvn_test = "mardia")
(mvn_result$multivariate_normality)
```

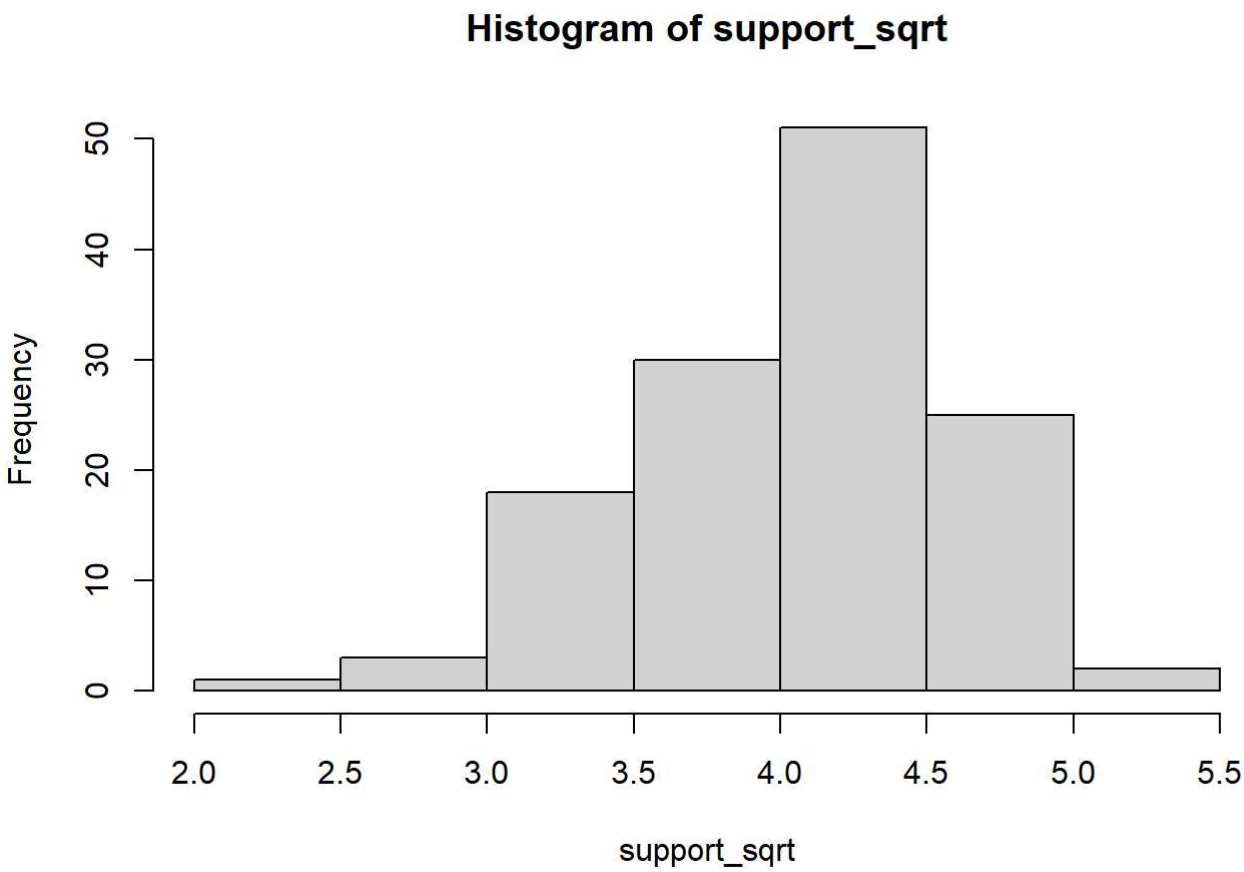
```
             Test Statistic p.value     Method      MVN
1 Mardia Skewness    36.517   0.398 asymptotic ✓ Normal
2 Mardia Kurtosis    -0.811   0.417 asymptotic ✓ Normal
```

```
mvn_result2 = mvn(data = features, mvn_test = "hz")
(mvn_result2$multivariate_normality)
```

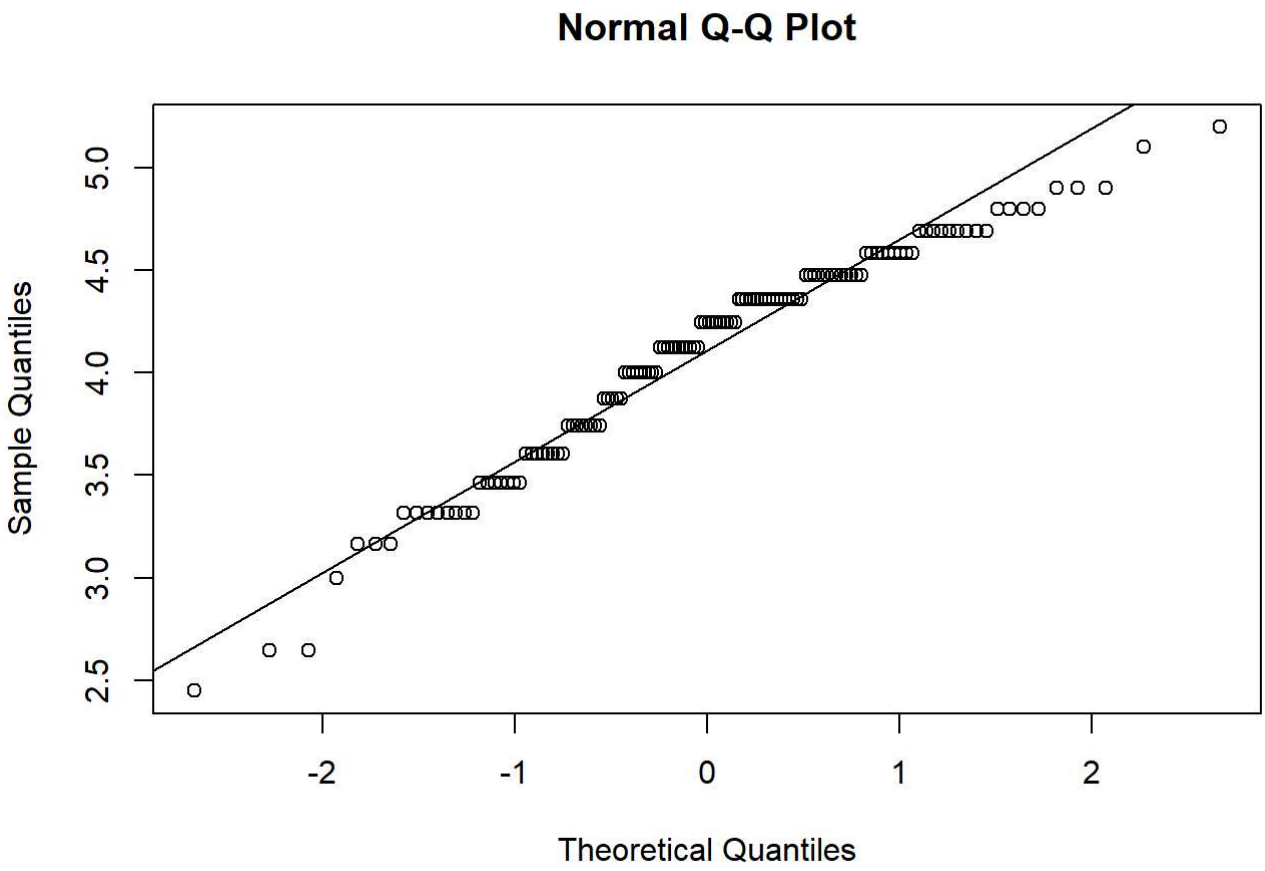
```
           Test Statistic p.value     Method      MVN
1 Henze-Zirkler     0.968   0.104 asymptotic ✓ Normal
```

c)

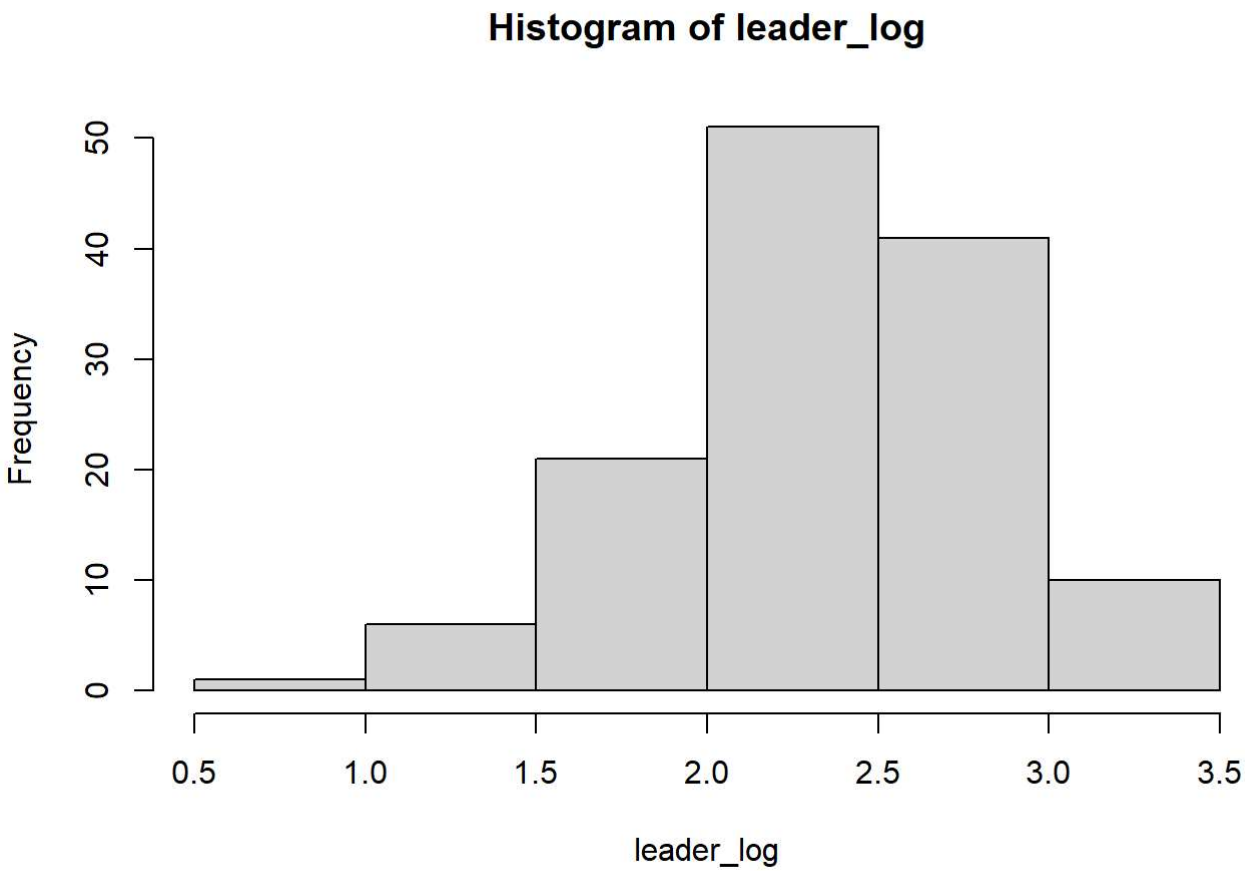
```
support_sqrt = sqrt(supp$V2)

hist(support_sqrt)
```



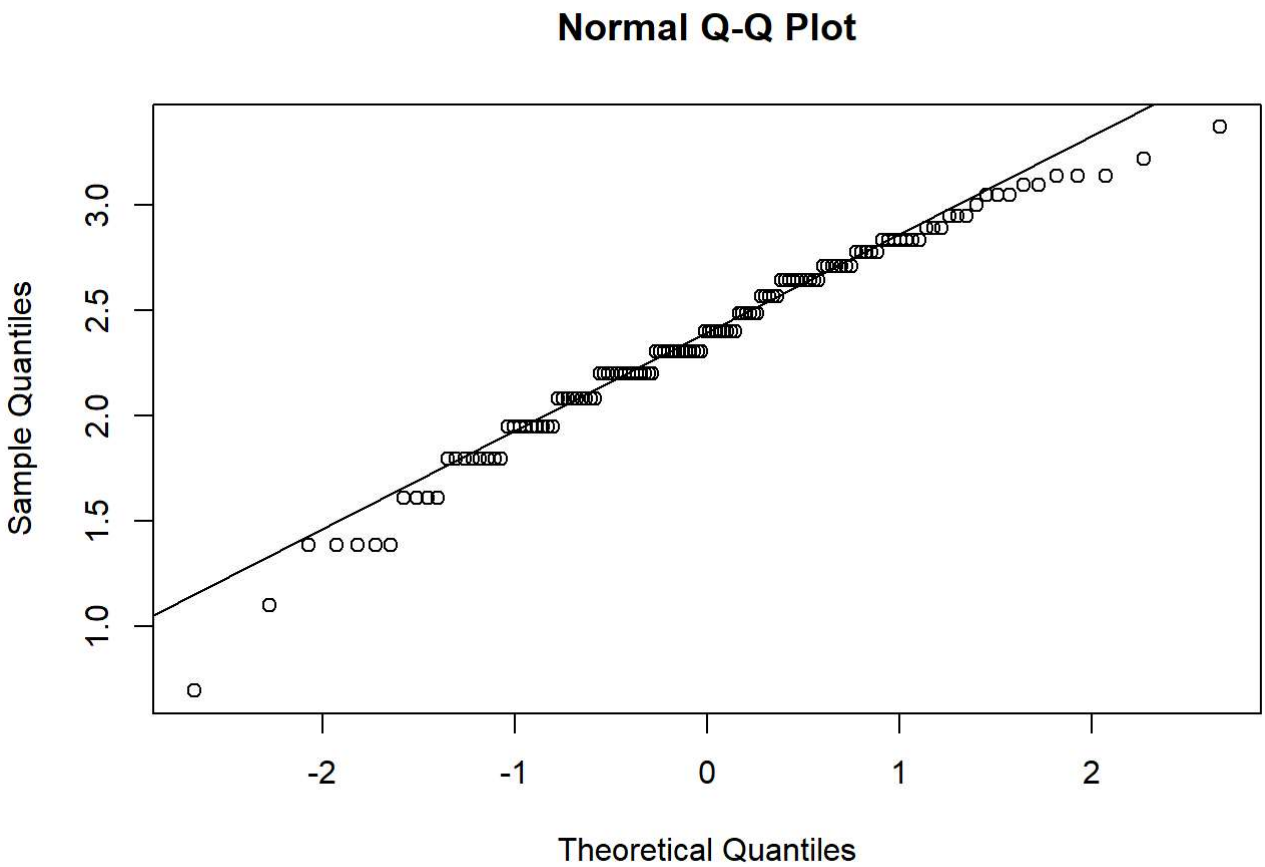

```
qqnorm(support_sqrt)
qqline(support_sqrt)
```




```
leader_log = log(leader$V5)

hist(leader_log)
```




```
qqnorm(leader_log)
qqline(leader_log)
```



Chapter 5

1

a)

```
n = 4
p = 2

X = matrix(c(2,12,
             8,9,
             6,9,
             8,10), byrow=T, nrow=4, ncol=2)
mu = c(7, 11)

xbar = colMeans(X)
Sinv = solve(cov(X))

T2 = n * t(xbar - mu) %*% Sinv %*% (xbar - mu)
(T2[1,1])
```

```
[1] 13.63636
```

b)

T^2 is distributed as $\frac{6}{2}F_{2,2} = 3F_{2,2}$ or $\frac{1}{3}T^2 \sim F_{2,2}$

c)

```
alpha = 0.05

fobs = (1/3) * T2[1,1]
fcrit = qf(0.05, 2, 2, lower.tail=F)

(fobs > fcrit)
```

```
[1] FALSE
```

Fail to reject $H_0 : \mu = [7, 11]$ at $\alpha = 0.05$. The data does not have enough evidence to conclude the mean differs from [7,11]

2)

```
C = matrix(c(1, -1,
             1, 1), byrow=T, nrow=2, ncol=2)
CX = t(C %*% t(X))
CXmu = C %*% c(7, 11)


CXbar = colMeans(CX)
CXSinv = solve(cov(CX))

CXT2 = n * t(CXbar - CXmu) %*% CXSinv %*% (CXbar - CXmu)
(CXT2[1,1])
```

```
[1] 13.63636
```

18)

```
data18 = read.table("../johnson_wichern_data/T5-2.DAT")
```

a)

```
mu0 = c(500, 50, 30)
alpha = 0.05
n = 87
p = 3

xbar18 = colMeans(data18)
Sinv18 = solve(cov(data18))
```

```
T2_18 = n * t(xbar18 - mu0) %*% Sinv18 %*% (xbar18 - mu0)

fobs = ((n-p)/(p*(n-1))) * T2_18[1,1]
fcrit = qf(0.05, p, n-p, lower.tail=F)

(fobs > fcrit)
```

```
[1] TRUE
```

Reject $H_0 : \mu = [500, 50, 30]$ at $\alpha = 0.05$. There is strong evidence that μ differs from [500, 50, 30]. This shows that the group of n=87 students scored <u>differently</u> than the group of thousands of college students which averaged [500, 50, 30].

b)

```
S18 = cov(data18)
eig18 = eigen(S18)

lengths = sqrt(eig18$values) * sqrt(((p*(n-1)) / (n*(n-p))) * fcrit)
directions = eig18$vectors

(lengths)
```

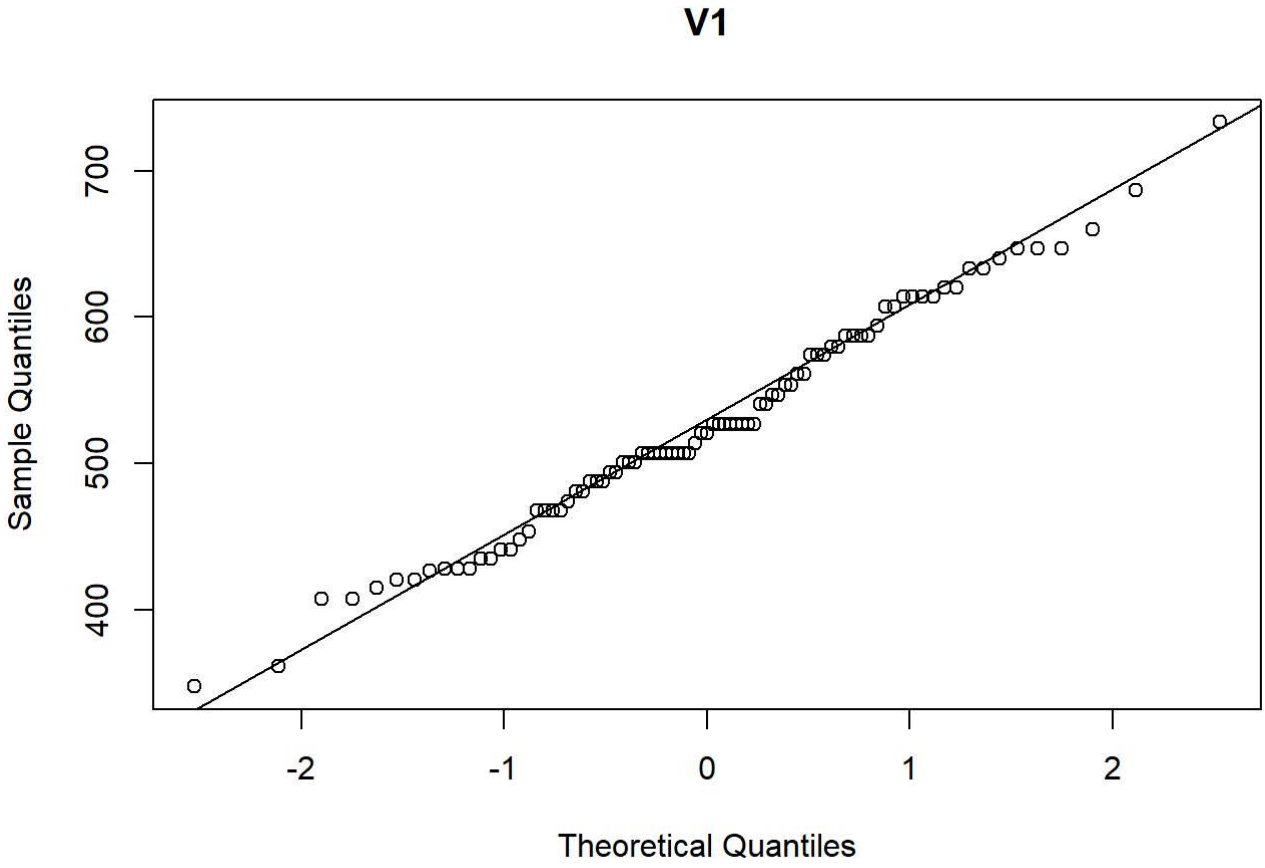
```
[1] 23.729998  2.472768  1.182500
```

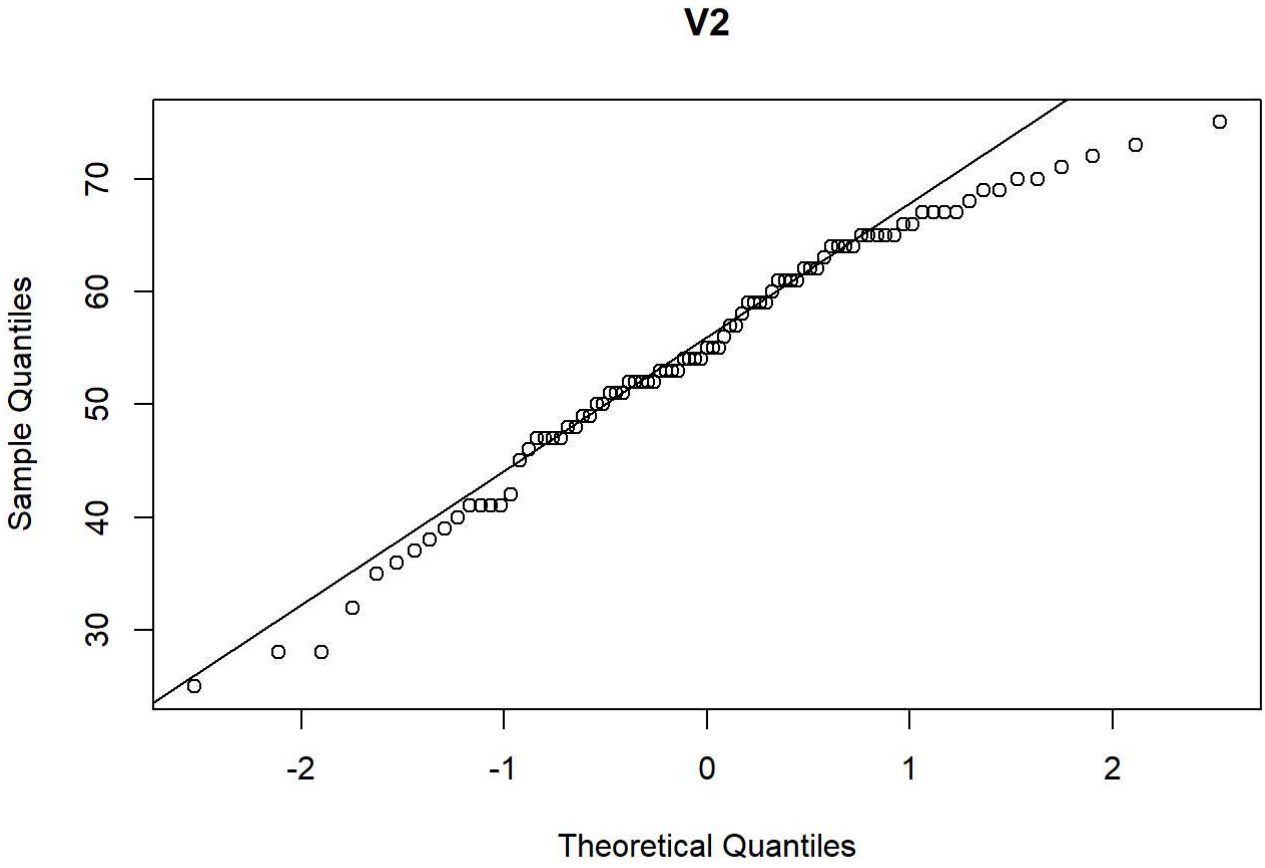
```
(directions)
```

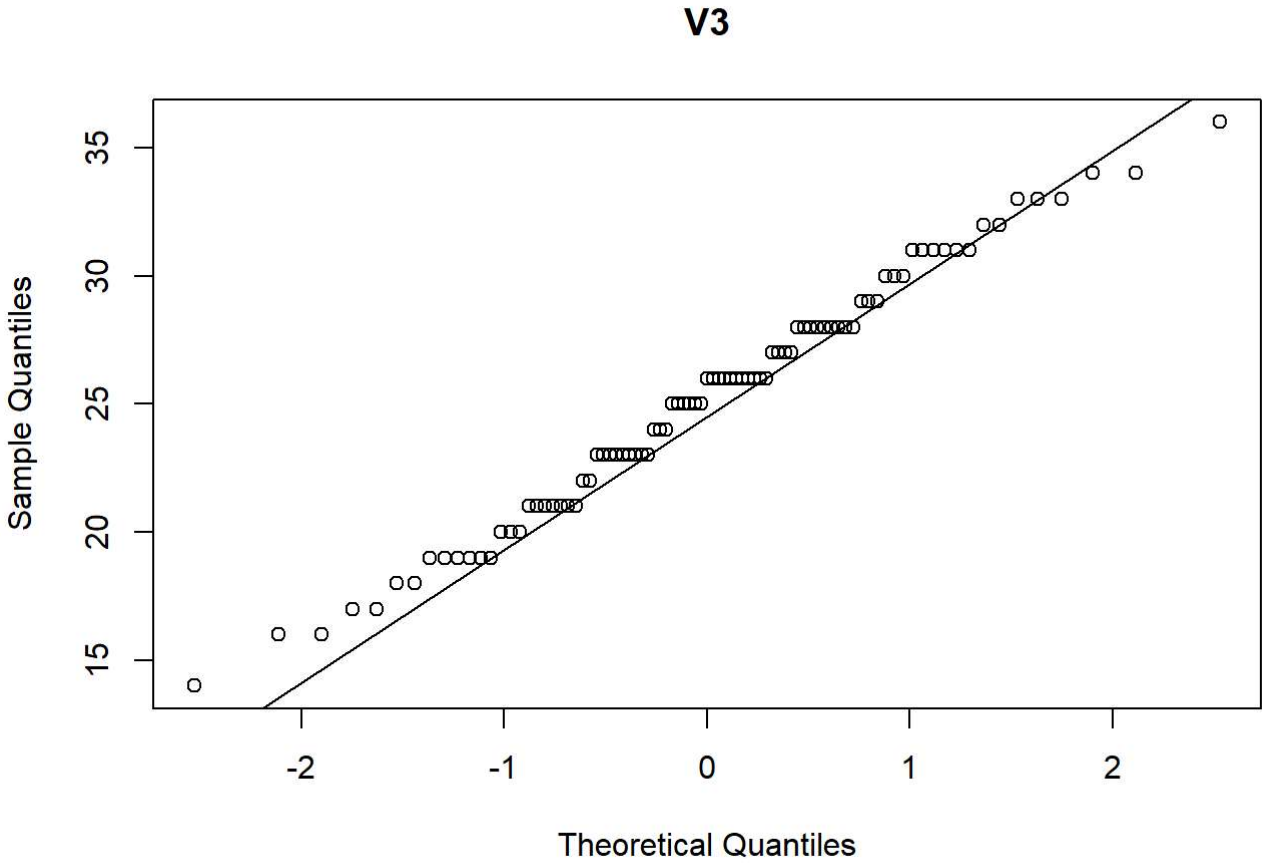
```
           [,1]         [,2]         [,3]
[1,] 0.99390539  0.103731534 -0.037307396
[2,] 0.10344339 -0.994589227 -0.009577815
[3,] 0.03809906 -0.005660238  0.999257936
```

c)

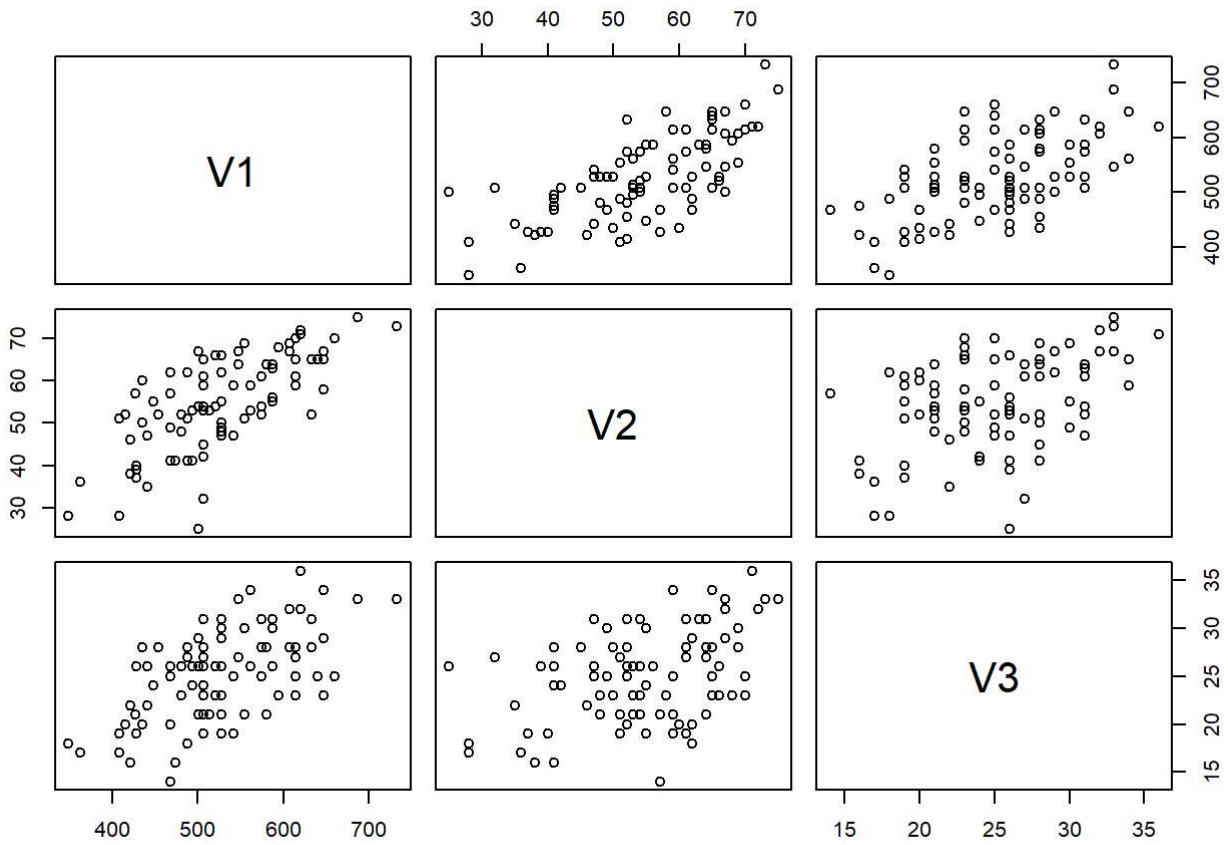
```
feature_names = c("V1", "V2", "V3")

for (i in 1:length(feature_names)){
  qqnorm(data18[[feature_names[i]]], main=feature_names[i])
  qqline(data18[[feature_names[i]]])
}
```


V1
Sample Quantiles
400
500
600
700
-2
-1
0
1
2
Theoretical Quantiles


V2
Sample Quantiles
30
40
50
60
70
-2
-1
0
1
2
Theoretical Quantiles




```
pairs(data18)
```



V1 and V3 appear to fall along the theoretical quantiles, whereas V2 has extremes on both ends of the distribution. Looking into the pair plots, I notice a positive covariance among many of the variables. V3 and V1 look the most normally distributed with one another.